

- 1 (ai) Internal energy is determined by the state of the system and it can be expressed as the sum of a random distribution of kinetic energy associated with the molecules of the system. [B1]

(ii) $pV = NkT$

$$(1.05 \times 10^5) \times (2.9 \times 10^{-4}) = N \times (1.38 \times 10^{-23}) \times (303) \quad [\text{M1}]$$

$$N = 7.282 \times 10^{21} = 7.28 \times 10^{21} \quad [\text{A1}]$$

(iii) $E_k = \frac{3}{2}kT$

$$= \frac{3}{2} \times (1.38 \times 10^{-23}) \times (303)$$

$$= 6.272 \times 10^{-21} = 6.27 \times 10^{-21} \text{ J} \quad [\text{A1}]$$

(b) (i) $\Delta U = \frac{3}{2}Nk\Delta T \quad [\text{M1}]$

$$= \frac{3}{2} (7.282 \times 10^{21}) (1.38 \times 10^{-23}) \times (357 - 303) \quad [\text{A1}]$$

$$= 8.14 \text{ J}$$

Allow e.c.f. from aii

- (ii) Since gas A undergoes an adiabatic compression, $Q = 0$.

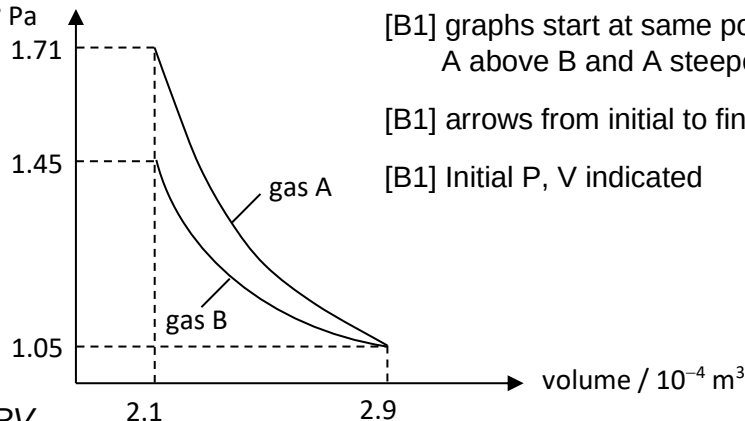
From the 1st law of thermodynamics,

$$\Delta U = Q + W$$

$$8.14 = 0 + W$$

$$W = 8.14 \text{ J} \quad [\text{A1}]$$

(iii) pressure / 10^5 Pa



[B1] graphs start at same point, with A above B and A steeper than B

[B1] arrows from initial to final state.

[B1] Initial P, V indicated

(For gas A, $\frac{PV}{T} = \text{constant}$.)

$$\frac{1.05 \times 10^5 \times 2.9 \times 10^{-4}}{303} = \frac{p_A \times 2.1 \times 10^{-4}}{357} \Rightarrow p_A = 1.71 \times 10^5 \text{ Pa}$$

For gas B, $PV = \text{constant}$.

$$(1.05 \times 10^5) \times (2.9 \times 10^{-4}) = p_B \times (2.1 \times 10^{-4}) \Rightarrow p_B = 1.45 \times 10^5 \text{ Pa}$$